

Unit V Applications and Advanced Topics

Unit V – Applications and Advanced Topics

1. SVM (Support Vector Machine)

SVM ek powerful supervised learning algorithm hai jo classification aur regression dono ke liye use hota hai. Ye maximum margin boundary dhoondhta hai jo classes ko best tareeke se separate kare.

Main Idea

- ? Best separating hyperplane find karna
- ? Support vectors important hote hain

Advantages

- ? Works well in high-dimensional space
- ? Useful for small and medium datasets

2. Bayesian Learning

Bayesian learning probabilistic reasoning par based hoti hai. Ye prior knowledge aur observed data ke basis par posterior probability compute karti hai.

Important Concepts

- ? Prior probability
- ? Likelihood
- ? Posterior probability

Usefulness

Ye uncertain situations aur probabilistic inference ke liye useful hai.

3. Machine Learning in Computer Vision

Machine learning ka use image classification, object detection, face recognition aur medical imaging mein hota hai.

Examples

- ? Face unlock
- ? Self-driving cars
- ? Medical scan analysis

4. Machine Learning in Speech Processing

Speech processing mein ML ka use speech recognition, speaker identification aur audio analysis mein hota hai.

Applications

- ? Voice assistants
- ? Speech-to-text systems
- ? Call center analytics

5. Machine Learning in NLP

Natural Language Processing mein ML ka use text classification, sentiment analysis, machine translation, chatbot development aur summarization mein hota hai.

Examples

- ? Chatbots
- ? Spam detection
- ? Language translation

6. ImageNet Case Study

ImageNet ek famous dataset hai jisme millions of labeled images thi. Is competition ne deep learning aur CNNs ki popularity ko aur zyada badhaya.

Significance

- ? Deep learning ka practical success dikha
- ? CNN models ko global recognition mili

7. Real-World Machine Learning Applications

Machine learning ab har field mein use ho rahi hai:

- ? Healthcare diagnosis
- ? Fraud detection
- ? Recommendation systems
- ? Weather forecasting
- ? Autonomous systems

8. Challenges in Real-World ML

Real-world systems mein ML ko implement karte waqt kuch challenges hote hain:

- ? Data quality issues
- ? Bias and fairness
- ? Privacy concerns
- ? Interpretability
- ? Computational cost

9. Future of Machine Learning

Machine learning ka future bohot bright hai. AI systems increasingly intelligent, adaptive aur autonomous ban rahe hain. Deep learning, transformers aur reinforcement learning abhi bhi rapidly grow kar rahe hain.

10. Important Exam Points

- ? SVM maximum margin boundary use karta hai.
- ? Bayesian learning probability-based hai.
- ? ML ka use computer vision, speech aur NLP mein hota hai.
- ? ImageNet case study deep learning ke growth ka important example hai.
- ? Real-world applications ke saath challenges bhi samjhe jaane chahiye.